

1. Uninformed Search (Blind Search)

2. Informed " (Heuristic ")

* mid-18-19

* Uninformed Search don't have the additional knowledge about search space

Some important information are not given (no additional info; no domain knowledge)

→ BFS (Breadth-First-Search)

→ DFS (Depth First Search)

→ Uniform Cost Search

→ Depth Limited Search

→ Iterative Deepening Search.

updated

updated

- * Properties of Searching Algorithms**
*** How to evaluate algorithms performance** mid-18th
- 1. Completeness:** It is said to be complete if it is guaranteed to find the solution if at least any solution exists at any random point.
 - 2. Optimality:** If a solution is found to be best solution (lowest-path cost) among all.
 - 3. Time complexity:** It is a measure of time for an algorithm to complete its task.
 - 4. Space Complexity:** How much memory is needed to perform the task.

* Comparing strategies:

uninformed search
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Criteria	BFS	US DFS	DFS	DLS	IDS	BS
Completeness	Yes ^a	Yes ^{a,b}	No	No	Yes ^a	Yes ^{a,d}
Time complexity	$O(b^d)$	$O[b^{(1+\frac{d}{e})}]$	$O(b^m)$	$O(b^l)$	$O(b^d)$	$O(b^{\frac{d}{2}})$
Space complexity	$O(b^d)$	$O[b^{1+\frac{d}{e}}]$	$O(b^m)$	$O(bd)$	$O(bd)$	$O(b^{\frac{d}{2}})$
Optimality	Yes ^c	Yes	No	No	Yes ^c	Yes ^{c,d}

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How to make heuristic function

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Here are some steps to create a heuristic function:

1. Understand the Problem
2. Define the state space
3. Determine the goal.
4. Choose a heuristic function.
common functions - manhattan distance, Euclidean distance
5. Admissibility: ensure that the function never overestimates the true cost to reach goal.
6. Fine-Tune and Test.
7. Integrate with ^{AI} search algorithm
8. Iterative Refinement.

Admissible heuristic function is

$$h(n) \leq h^*(n)$$

$h(n)$ = heuristic cost

$h^*(n)$ = estimated "

problems.
 * Define and Illustrate admissible and consistent heuristic for estimating the optimality of A* search.

Answer:

An admissible heuristic is one that never overestimate the cost to reach the goal. Because $g(n)$ is actual cost to reach n along current path and $f(n) = g(n) + h(n)$, we can say that $f(n)$ never overestimate the true

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An obvious example of an admissible heuristic is straight line distance. ~~hsl~~. ~~then~~ suppose we want to go Bucharest from Arad. Straight line is admissible because the shortest path between any two points is a straight line, so straight line can't be an overestimate. We can see the example of A* tree for Bucharest. The values of g are computed from step cost and ~~hsl~~ are given. Notice here, Bucharest first appears on the frontier at step (e) but ~~it~~ it is not selected for expansion because its $fcost(450)$ is higher than that of Pitesti(417). And we know to be admissible the function needs to be $f(n) \leq g(n) + h(n)$ and $h(n) \leq h^*(n)$.

As Pitesit 414 has low cost as 417
so, algorithm will not settle for a
solution that cost 450.

Second condition is consistency

It is required only for A* search.

A heuristic $h(n)$ is consistent if

for every node n and successor

n' of n generated by any action

the estimated cost of reaching

goal n no longer greater than the

step cost of getting to n' plus

the estimated cost of reaching goal

from n' .

$$h(n) \leq c(n, a, n') + h(n')$$

This is a form of triangle inequality.

For example, Let's consider the goal

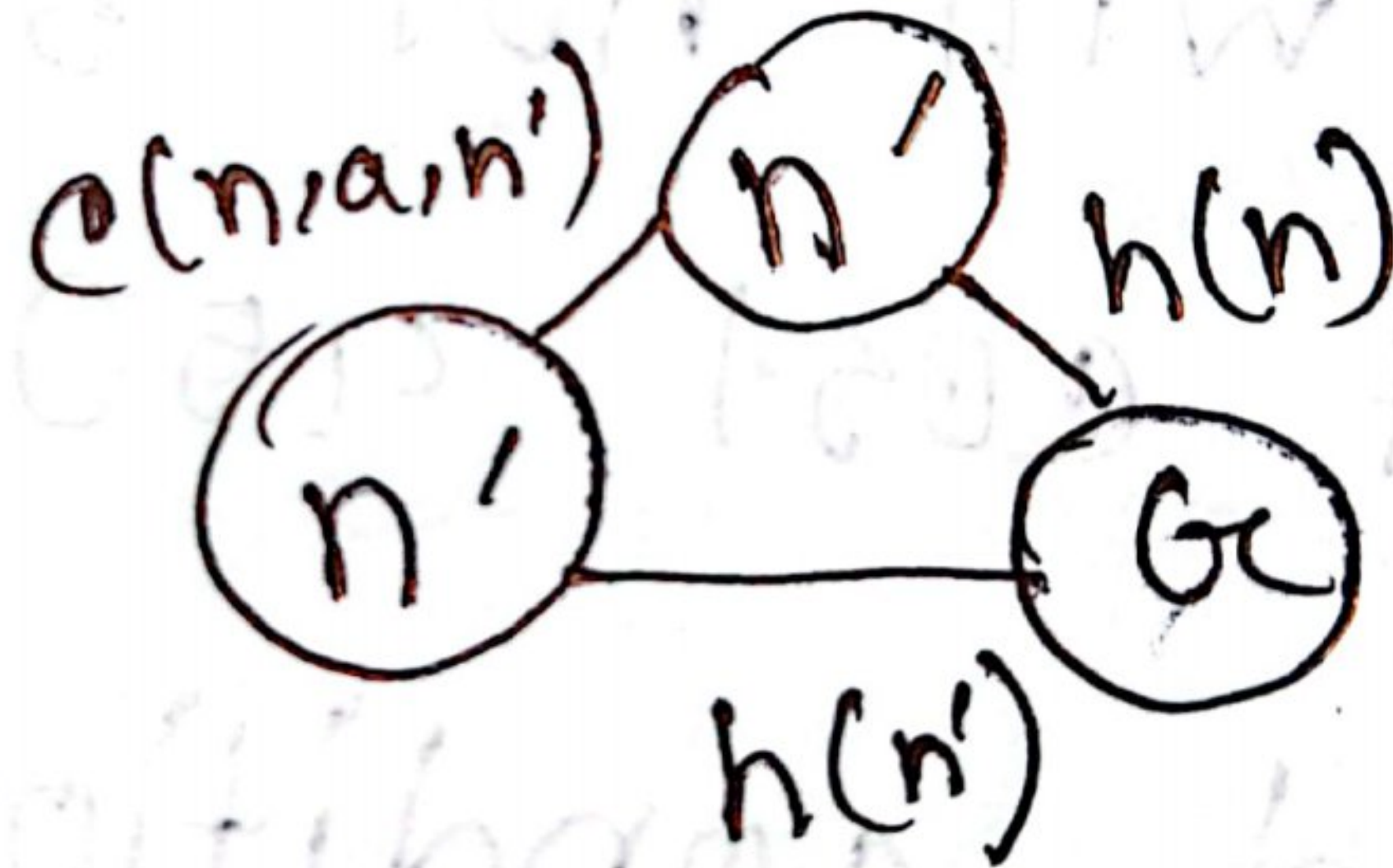
is to reach G from node S.

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$S - A - B - C - D - G$



properties of A^* search:

completeness: Yes, it is complete

* Heuristic may be:

1. $h(n) > \text{actual cost}$ (optimal solution is not possible)
2. $h(n) = \text{actual cost}$ (best case scenario)
3. $h(n) < \text{actual cost}$ (admissible, consistent heuristic)

optimality of A* search algorithm.

knowledge after observation and take actions.

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*Algorithm of knowledge based agent:

function KB-AGENT(percept) returns an action

KB: Knowledge base

t: time (counter initially 0)

TELL (KB, MAKE-PERCEPT-SENTENCE(
percept, t))

action = ASK (KB, MAKE-ACTION-QUERY(t))

TELL (KB, MAKE-ACTION-SENTENCE(action, t))

t = t + 1

return action.

* Prove that wumpus world is in the room (1,3) : mid-18-19-4

⇒ Apply Modus Ponens with $\neg S_{11}$ and R_1 :

$$\neg S_{11} \rightarrow \neg W_{11} \wedge \neg W_{12} \wedge \neg W_{21}$$

$$\neg S_{11}$$

$$\neg W_{11} \wedge \neg W_{12} \wedge \neg W_{21}$$

⇒ Apply And Elimination Rule:
After applying And elimination we get : $\neg W_{11}, \neg W_{12}, \neg W_{21}$

⇒ Apply Modus Ponens to $\neg S_{21}$ and R_2 :

$$\neg S_{21} \rightarrow \neg W_{21} \wedge \neg W_{22} \wedge \neg W_{31}$$

$$\neg S_{21}$$

$$\neg W_{21} \wedge \neg W_{22} \wedge \neg W_{31}$$

⇒ Apply And Elimination rule!
after applying And elimination we
get, $\neg W_{21}, \neg W_{22}, \neg W_{31}$

⇒ Apply MP to S_{12} and R_4 °

$S_{12} \rightarrow W_{13} \vee W_{12} \vee W_{22} \vee W_{11}$

S_{12}

$W_{13} \vee W_{12} \vee W_{22} \vee W_{11}$

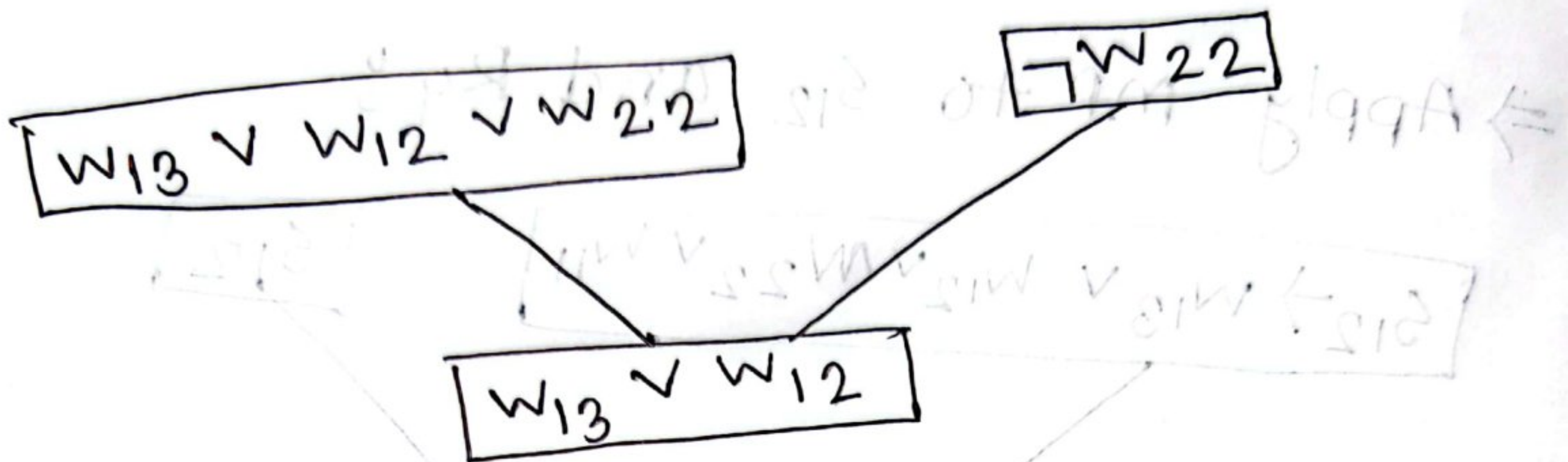
⇒ Apply Unit resolution on $W_{13} \vee W_{12} \vee$
 $W_{22} \vee W_{11}$ and $\neg W_{11}$ °

$W_{13} \vee W_{12} \vee W_{22} \vee W_{11}$

$\neg W_{11}$

$W_{13} \vee W_{12} \vee W_{22}$

⇒ Apply Unit resolution on $w_{13} \vee w_{12} \vee w_{22}$
 and $\neg w_{22}$:



⇒ Apply Unit resolution on $w_{13} \vee w_{12}$
 and $\neg w_{12}$:

